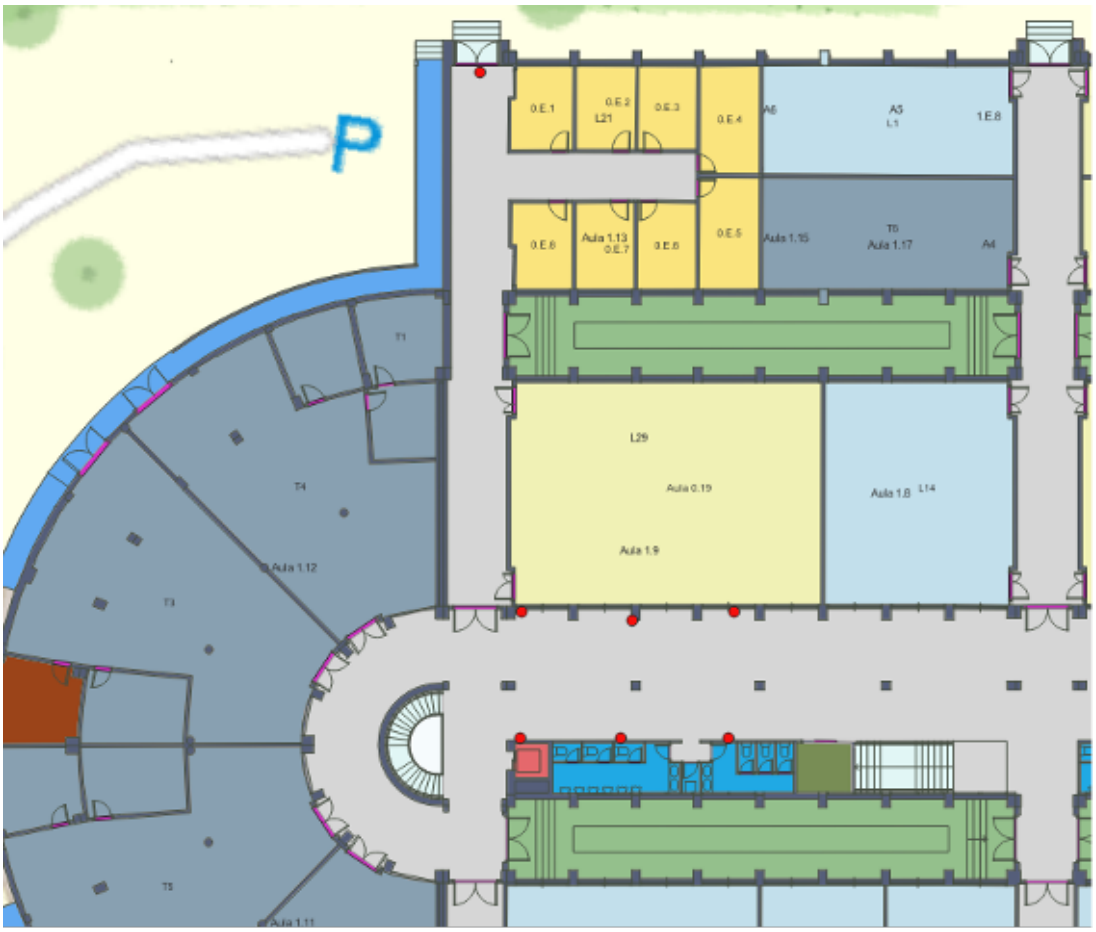

Indoor BLE Localization Dataset

Technical Report: Experimental Environment Description

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Abstract

This technical report documents the indoor experimental environment used to collect the Bluetooth Low Energy (BLE) RSSI fingerprint dataset for indoor localization experiments. The report describes the physical characteristics of the measurement areas, the deployment of BLE beacons, the definition of spatial sectors, and the acquisition setup used during the data collection campaign. This document complements the dataset repository and the associated data article by providing additional information about the measurement environment to facilitate interpretation and reuse of the released data.

1 Introduction

Indoor localization based on Bluetooth Low Energy (BLE) signals has become a widely studied problem due to the widespread availability of BLE-capable smartphones and low-cost beacon devices. RSSI fingerprinting approaches are commonly used to estimate the location of a receiver by comparing measured signal strengths with previously collected reference measurements.

This technical report provides a detailed description of the experimental environment used to acquire a BLE RSSI fingerprint dataset collected in a university building. The purpose of this document is to document the physical environment, beacon deployment configuration, sector layout, and acquisition procedure used during the measurement campaign.

The report complements the public dataset repository and the associated data publication by providing additional context about the measurement setup and environment.

2 Experimental Environment Overview

The measurement campaign was conducted inside the Escuela Superior de Ingeniería Informática de Albacete (ESIIAB), University of Castilla-La Mancha, Spain.

Two indoor areas of the building were used during the acquisition process:

- a corridor area located on the first floor
- a larger open area located on the ground floor

These environments represent realistic indoor conditions for BLE signal propagation, including walls, structural columns, metallic elements, furniture, and occasional human activity.

3 Experimental Environment Overview

The dataset was collected in two indoor areas of the Escuela Superior de Ingeniería Informática de Albacete (ESIIAB), University of Castilla-La Mancha, Spain. Measurements were acquired both on the first floor and on the ground floor of the building.

Table 1 summarizes the main characteristics of the released dataset.

Table 1: Summary of the BLE RSSI fingerprint dataset.

Property	Value
Number of BLE beacons	6
Beacon model	MOKO SMART H2
Receiver device	Xiaomi Mi 9T Pro smartphone
Number of floors	2
Number of sectors	28 (7 first floor, 21 ground floor)
Transmission power levels	3 (-8, -4, 0 dBm)
Features per sample	6 RSSI values (B1–B6)
Target variable	Sector label
Dataset files released	9 CSV files
Measurement environment	University indoor building (ESIIAB)

The selected areas provide relatively open indoor environments suitable for BLE signal measurements. The first-floor corridor includes a metallic safety railing as its main structural element, while the ground-floor area contains reinforced concrete columns and tables with metallic frames. These elements introduce realistic propagation conditions for Bluetooth Low Energy signals.

Both measurement areas have a length of approximately 10.5 meters. The first-floor corridor has a width of approximately 2.4 meters, while the ground-floor area has a width of approximately 6 meters.

During the acquisition process, the smartphone receiver was positioned approximately at the center of each sector while recording RSSI measurements.

4 First-Floor Measurement Area

The first-floor measurement area consists of an indoor corridor approximately 10.5 m long and 2.4 m wide. This corridor was divided into seven rectangular sectors labelled `1A` to `7A`. Each sector measures approximately 1.2×2.4 m. Adjacent sectors are separated by longitudinal gaps of approximately 0.4 meters.

Figure 1: Sector layout and beacon deployment on the first floor.

(a) View from sector 1A (b) View from sector 7A

Figure 2: Photographs of the first-floor measurement corridor.

5 Ground-Floor Measurement Area

The ground-floor measurement area corresponds to a larger open indoor space approximately 10.5 m long and 6 m wide.

This area was divided into twenty-one sectors labelled `1B` to `21B`. The sectors are arranged in a 3×7 grid layout. Most sectors measure approximately 1.2×1.2 m. The sectors vertically aligned with the first-floor sectors (`1B`, `4B`, `7B`, `10B`, `13B`, `16B`, and `19B`) have dimensions of 1.2×2.4 meters.

Figure 3: Sector layout and beacon deployment on the ground floor.

(a) View from sectors 1B–3B

(b) View from sectors 19B–21B

Figure 4: Photographs of the ground-floor measurement environment.

6 BLE Beacon Deployment

Six BLE navigation beacons (MOKO SMART H2) were used during the measurement campaign and labelled `B1` to `B6`.

The beacons were deployed in a rectangular configuration across the two measurement areas. Beacons `B1`, `B2`, and `B3` were positioned on the side closest to the first-floor sectors, while beacons `B4`, `B5`, and `B6` were positioned on the opposite side.

All beacons were installed at an approximate height of 6 meters above the floor level. Adjacent beacons were separated by approximately 6 meters. The opposing beacon pairs (`B1`–`B4`, `B2`–`B5`, and `B3`–`B6`) were also approximately 6 meters apart.

This rectangular deployment geometry provides spatial coverage across the measurement sectors.

It is important to note that the experimental areas are frequently used by students and faculty members. Consequently, occasional pedestrian movement and seated individuals were present during some measurement sessions, introducing realistic variations in BLE signal propagation.

7 Sector Definition

The dataset uses sector labels as target classes for indoor localization experiments.

The sectors are defined as follows:

- First floor: sectors `1A` to `7A`
- Ground floor: sectors `1B` to `21B`

These sector labels correspond to the spatial regions shown in the floor maps presented earlier in this report.

8 Devices and Software Tools

This section describes the main hardware and software elements used during the BLE RSSI data acquisition campaign.

8.1 BLE Beacons

The measurement campaign used six MOKO SMART H2 indoor navigation beacons. These devices operate in the 2.4 GHz BLE band and support beacon-based communication protocols such as iBeacon and Eddystone. Each device broadcasts its own MAC address and the corresponding Bluetooth signal strength, which was used in this work as the RSSI source for indoor localization measurements.

The six deployed beacons were labelled `B1` to `B6` for identification during acquisition and dataset construction.

8.2 Receiver Smartphone

A Xiaomi Mi 9T Pro smartphone was used as the receiver device during the acquisition campaign. The device supports Bluetooth 5.0 and was used both to configure the BLE beacons and to record RSSI measurements from the six deployed devices.

During the campaign, the smartphone was positioned approximately at the center of each labeled sector while measurements were being recorded.

8.3 Beacon Configuration Software

Beacon parameters were configured using the `BeaconX Pro` mobile application provided by MOKO SMART. This application was used to detect the beacons, verify their MAC addresses, and set the transmission power level used during each acquisition campaign.

8.4 BLE Logging Software

RSSI measurements were recorded using the `BLE Scanner and Logger` mobile application developed by THINGSUP. This application was used to scan the BLE devices, filter the target beacons, and export the recorded measurements as CSV files.

For each sector and transmission power level, the logging process was repeated until approximately 100 RSSI samples were obtained from the six beacons.

9 Data Acquisition Setup

RSSI measurements were collected using a Xiaomi Mi 9T Pro smartphone acting as the receiver device. The smartphone was used both to configure the BLE beacons and to record the RSSI measurements.

Beacon configuration was performed using the `BeaconX Pro` mobile application provided by MOKO SMART. The application was used to detect the beacons, verify their MAC addresses, and configure the transmission power levels used during the acquisition campaigns.

RSSI measurements were recorded using the `BLE Scanner and Logger` mobile application developed by THINGSUP. The application was used to scan the BLE devices, filter the target beacons, and export the recorded measurements as CSV files.

Three transmission power levels were used during the acquisition campaign:

- -8 dBm
- -4 dBm
- 0 dBm

For each transmission power level and each sector, the receiver smartphone was positioned approximately at the center of the sector while recording RSSI measurements from the six deployed beacons. During each logging session the objective was to capture approximately 100 RSSI samples per sector.

The measurement procedure followed the same workflow for all sectors:

1. Verify that the six target beacons are active and detectable through a BLE scan.
2. Filter the detected devices to include only the target beacons using their device names and MAC addresses.
3. Start the logging process using the BLE Scanner and Logger application.
4. Record RSSI measurements for the required duration until approximately 100 samples are collected.
5. Export the measurements as CSV files for subsequent processing.

The acquisition campaign was conducted separately for each transmission power level. After completing all sector measurements for one power level, the transmission power of the six beacons was updated and the measurement procedure was repeated.

Measurements on the first floor were conducted during three afternoon sessions with a single operator positioned at the center of each of the seven sectors. Measurements on the ground floor required approximately one week due to the larger number of sectors. As in the first-floor campaign, a single operator performed the measurements.

Although the measurements were mainly performed under low-occupancy conditions, occasional pedestrians and seated individuals were present in parts of the ground-floor area. In some cases measurements were repeated in order to obtain samples under both occupied and non-occupied conditions.

10 Baseline Benchmark Results

To provide a reference point for potential dataset users, a set of baseline machine learning experiments was performed using the released datasets.

Several standard classification algorithms were evaluated for predicting the sector label from RSSI measurements, including Random Forest, Extra Trees, and Gradient Boosting models.

Table 2 summarizes the best observed results obtained for each transmission power level and dataset partition. Each entry reports the highest classification accuracy observed among the evaluated models together with the corresponding mean positioning error (EPM). The algorithm producing the best result is indicated in parentheses.

Table 2: Baseline benchmark results. Each entry reports the best observed classification accuracy (%) and the corresponding mean positioning error (meters). The algorithm producing the best result is indicated in parentheses. ET: Extra Trees. GBM: Gradient Boosting Machine.

Tx (dBm)	First Floor	Ground Floor	All Floors
-8	73.29% / 0.775 m (ET)	64.95% / 0.621 m (GBM)	62.68% / 0.523 m (ET)
-4	73.43% / 0.820 m (ET)	58.38% / 0.426 m (ET)	58.00% / 0.551 m (ET)
0	70.86% / 0.635 m (ET)	59.71% / 0.672 m (ET)	59.04% / 0.463 m (ET)

11 Dataset Repository Structure

The dataset repository is organized as follows:

```
README.md
data/
metadata/
images/
documentation/
```

Each dataset file contains six RSSI features corresponding to the six BLE beacons and one target column representing the sector label.